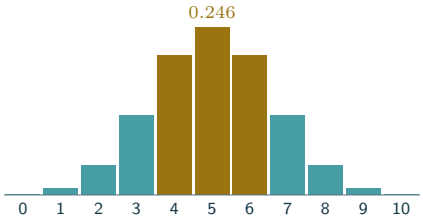
PROBABILITY LADDER 6 OF 9

The binomial distribution

Count the successes in a fixed number of independent trials, and the probabilities come from binomial coefficients.

1 / THE RULE AND THE PICTURE

In n independent trials with success probability p, the chance of k successes is $\binom{n}{k}p^k(1-p)^{n-k}$



heads in 10 fair tosses, with 4, 5 or 6 in gold

Each particular sequence with k successes has probability $p^k(1-p)^{n-k}$, and there are $\binom{n}{k}$ such sequences, so

$$P(X=k)=\binom{n}{k}p^k(1-p)^{n-k},\quad E[X]=np,\quad \operatorname{Var}X=np(1-p)$$

For ten fair tosses $\binom{10}{5}/1024=\frac{252}{1024}=0.246$ is the most likely count.

2 / WORKED EXAMPLES

Exactly three sixes in ten rolls of a die

Here $n=10$, $p=\frac{1}{6}$ and $k=3$, so the probability is $\binom{10}{3}\left(\frac{1}{6}\right)^3\left(\frac{5}{6}\right)^7=120\cdot\frac{1}{216}\cdot 0.279=0.155$.
For at least one success the complement is quickest: at least one defect among ten items that are each defective with probability 0.1 has probability $1-0.9^{10}=0.651$.

3 / THE DISTRIBUTION FOR TEN FAIR TOSSES

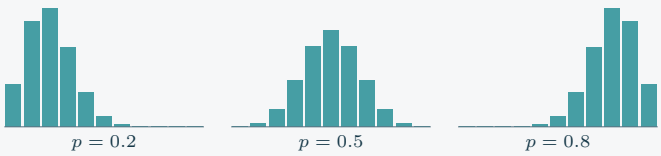
The probabilities are the row of Pascal's triangle divided by $2^{10}=1024$

k	$\binom{10}{k}$	probability	decimal
0	1	1/1024	0.0010
1	10	10/1024	0.0098
2	45	45/1024	0.0439
3	120	120/1024	0.1172
4	210	210/1024	0.2051
5	252	252/1024	0.2461

The distribution is symmetric about 5.
The chance of 4, 5 or 6 heads is $\frac{210+252+210}{1024}=\frac{672}{1024}=0.656$, and the chance of two or fewer is $\frac{1+10+45}{1024}=\frac{56}{1024}=0.055$.

4 / THE EFFECT OF p

Small p piles the probability at the left, large p at the right, and $p=\frac{1}{2}$ is symmetric



The means are $np=2,\ 5,\ 8$ and the variance $np(1-p)=1.6,\ 2.5,\ 1.6$, so the spread is greatest for $p=\frac{1}{2}$. For $n=20$ and $p=0.3$ the mean is 6, the standard deviation $\sqrt{4.2}=2.05$, and the most likely count 6 has probability 0.192.

5 / WHY np AND $np(1-p)$ / THE GALTON BOARD

A count of successes is a sum of n independent 0-1 trials

Each trial contributes mean p and variance $p(1-p)$, and means and independent variances add, giving np and $np(1-p)$.
In a Galton board with 8 rows each peg sends the ball left or right with probability $\frac{1}{2}$, so the final bins have probabilities $1, 8, 28, 56, 70, 56, 28, 8, 1$ out of 256, the binomial row with $n=8$.

ASIDE / INDEPENDENCE IS ESSENTIAL

Drawing cards without replacement is not binomial, because the success probability changes after each draw.

LINKS TO OTHER SHEETS

The coefficients come from *Counting: arrangements and choices*, and the mean np from *Random variables, expectation and variance*; the next sheet is *Waiting times: the geometric and Poisson distributions*.

Convention. X counts successes; n is the number of trials; p the success probability; trials are independent with equal p.